

DEFINE & TEST

JSS 1-3 | Applied Systems Garden

Use scientific investigation to define a question about whole-system relationships. Quantify the response and defend a recommendation.

MISSION KIT system map, records, calculator

EVIDENCE stock, flow, feedback, trade-off or scenario evidence

GM-391 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Define with system map, records, calculator. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for stock, flow, feedback, trade-off or scenario evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Define system boundary. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Applied Systems Garden, including the evidence that triggers the decision.

APPLIED SCIENCE MISSION

JSS 1-3 | Applied Systems Garden

Use scientific investigation to hypothesize a question about whole-system relationships. Quantify the response and defend a recommendation.

MISSION KIT system map, records, calculator

EVIDENCE stock, flow, feedback, trade-off or scenario evidence

GM-392 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Hypothesize with system map, records, calculator. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for stock, flow, feedback, trade-off or scenario evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Define system boundary. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Applied Systems Garden, including the evidence that triggers the decision.

DESIGN THE EVIDENCE

JSS 1-3 | Applied Systems Garden

Use scientific investigation to design a question about whole-system relationships. Quantify the response and defend a recommendation.

MISSION KIT system map, records, calculator

EVIDENCE stock, flow, feedback, trade-off or scenario evidence

GM-393 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Design with system map, records, calculator. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for stock, flow, feedback, trade-off or scenario evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Define system boundary. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

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REPLICATE & TEST

JSS 1-3 | Applied Systems Garden

Use scientific investigation to replicate a question about whole-system relationships. Quantify the response and defend a recommendation.

MISSION KIT system map, records, calculator

EVIDENCE stock, flow, feedback, trade-off or scenario evidence

GM-394 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Replicate with system map, records, calculator. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for stock, flow, feedback, trade-off or scenario evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Define system boundary. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Applied Systems Garden, including the evidence that triggers the decision.

APPLIED SCIENCE MISSION

JSS 1-3 | Applied Systems Garden

Use scientific investigation to quantify a question about whole-system relationships. Quantify the response and defend a recommendation.

MISSION KIT system map, records, calculator

EVIDENCE stock, flow, feedback, trade-off or scenario evidence

GM-395 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Quantify with system map, records, calculator. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for stock, flow, feedback, trade-off or scenario evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Define system boundary. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Applied Systems Garden, including the evidence that triggers the decision.

ANALYZE THE EVIDENCE

JSS 1-3 | Applied Systems Garden

Use scientific investigation to analyze a question about whole-system relationships. Quantify the response and defend a recommendation.

MISSION KIT system map, records, calculator

EVIDENCE stock, flow, feedback, trade-off or scenario evidence

GM-396 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Analyze with system map, records, calculator. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for stock, flow, feedback, trade-off or scenario evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Define system boundary. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Applied Systems Garden, including the evidence that triggers the decision.

DIAGNOSE & TEST

JSS 1-3 | Applied Systems Garden

Use scientific investigation to diagnose a question about whole-system relationships. Quantify the response and defend a recommendation.

MISSION KIT system map, records, calculator

EVIDENCE stock, flow, feedback, trade-off or scenario evidence

GM-397 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Diagnose with system map, records, calculator. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for stock, flow, feedback, trade-off or scenario evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Define system boundary. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Applied Systems Garden, including the evidence that triggers the decision.

APPLIED SCIENCE MISSION

JSS 1-3 | Applied Systems Garden

Use scientific investigation to model a question about whole-system relationships. Quantify the response and defend a recommendation.

MISSION KIT system map, records, calculator

EVIDENCE stock, flow, feedback, trade-off or scenario evidence

GM-398 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Model with system map, records, calculator. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for stock, flow, feedback, trade-off or scenario evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Define system boundary. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Applied Systems Garden, including the evidence that triggers the decision.

EVALUATE THE EVIDENCE

JSS 1-3 | Applied Systems Garden

Use scientific investigation to evaluate a question about whole-system relationships. Quantify the response and defend a recommendation.

MISSION KIT system map, records, calculator

EVIDENCE stock, flow, feedback, trade-off or scenario evidence

GM-399 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Evaluate with system map, records, calculator. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for stock, flow, feedback, trade-off or scenario evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Define system boundary. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Applied Systems Garden, including the evidence that triggers the decision.

RECOMMEND & TEST

JSS 1-3 | Applied Systems Garden

Use scientific investigation to recommend a question about whole-system relationships. Quantify the response and defend a recommendation.

MISSION KIT system map, records, calculator

EVIDENCE stock, flow, feedback, trade-off or scenario evidence

GM-400 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Recommend with system map, records, calculator. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for stock, flow, feedback, trade-off or scenario evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Define system boundary. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Applied Systems Garden, including the evidence that triggers the decision.